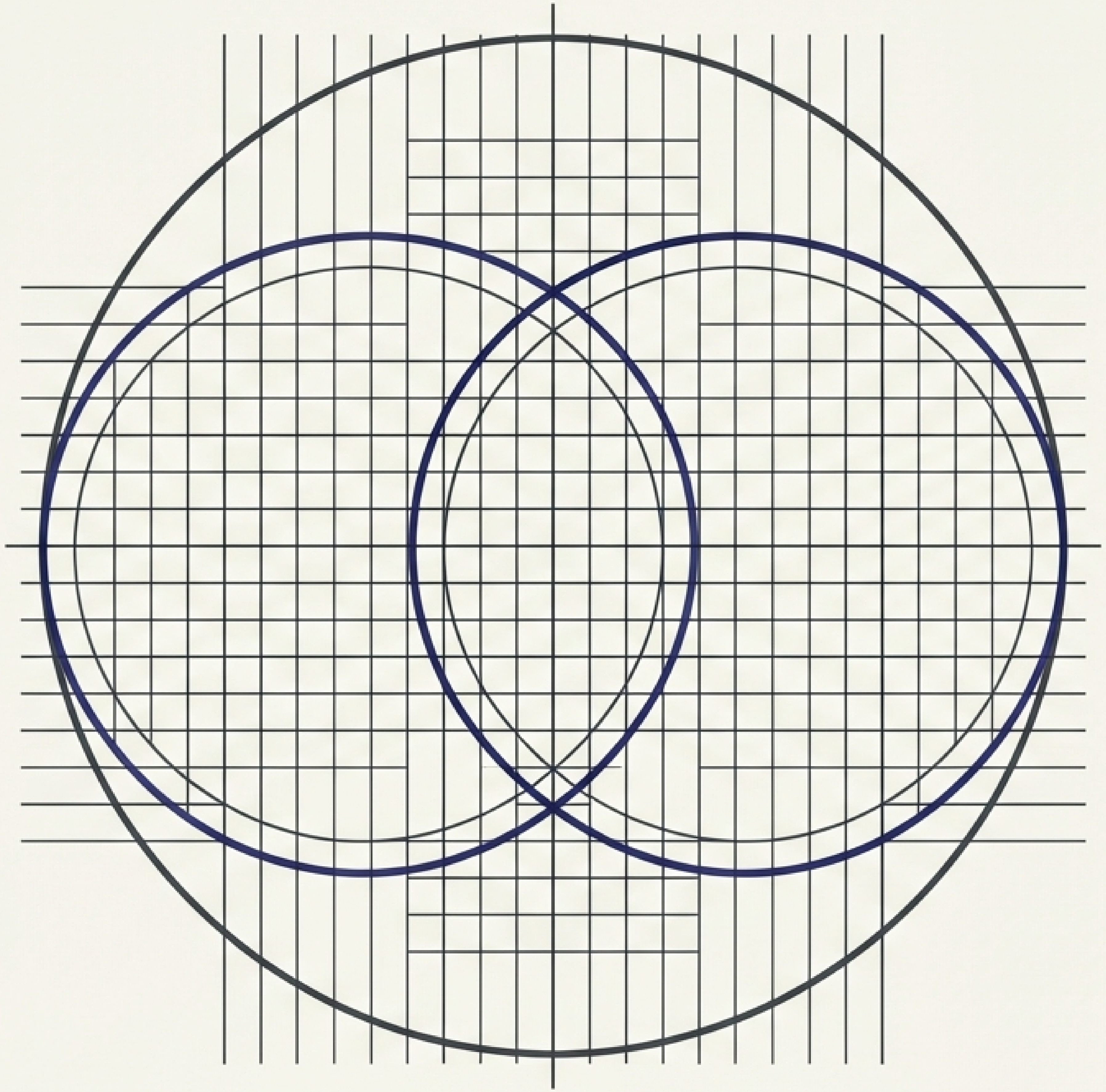

Decoding Poetic Geometry

Distributional Semantics, Embedding Geometries, and Sequence Classification

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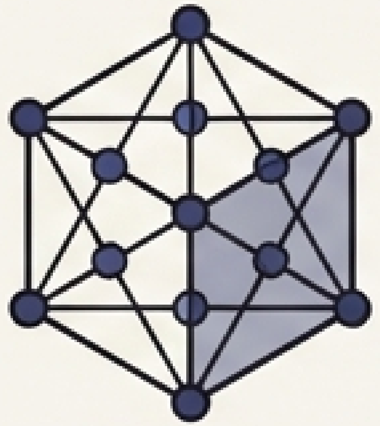
An End-to-End Analysis Pipeline

We transform raw poetic text into actionable semantic insights, evaluating the precise point where dense neural embeddings diverge from sparse matrix representations.

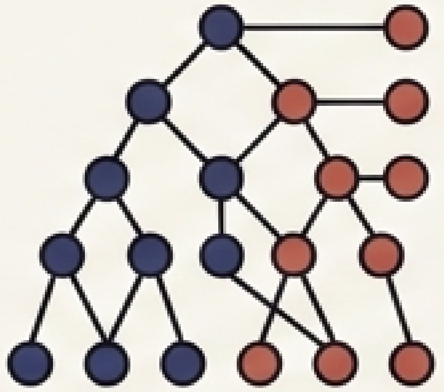
Corpus

Raw poetry to structured matrices.



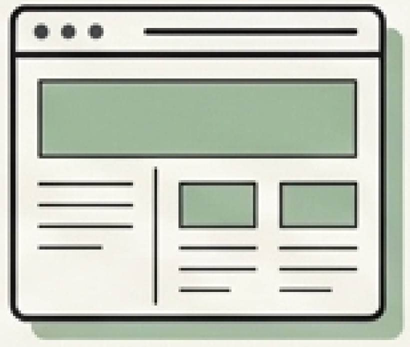
Vectors

Training Word2Vec & GloVe semantic spaces.



Classification

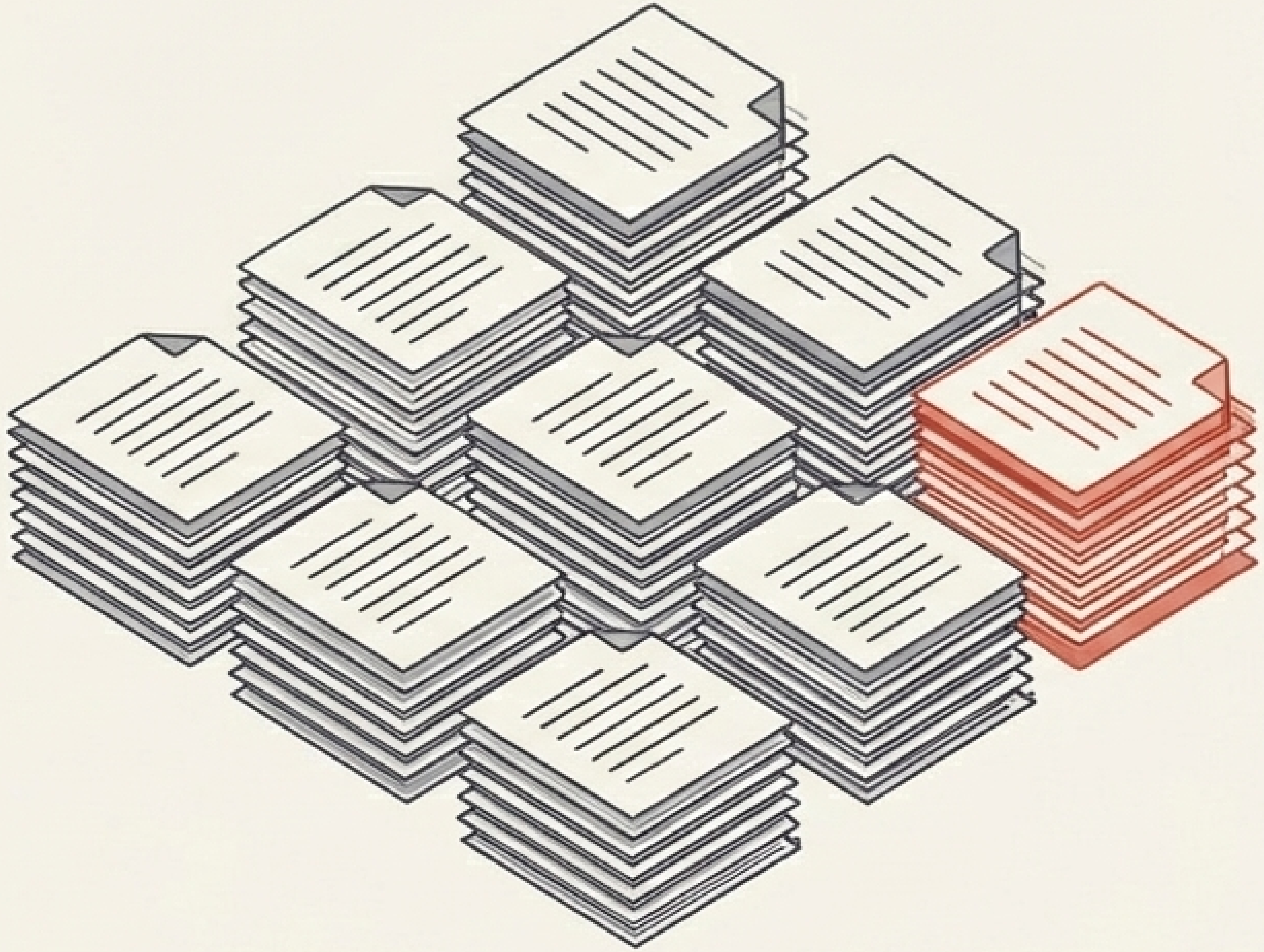
Author prediction via recurrent neural architectures.



Dashboard

Interactive FastAPI integration layer.

The Poetic Corpus Expansion

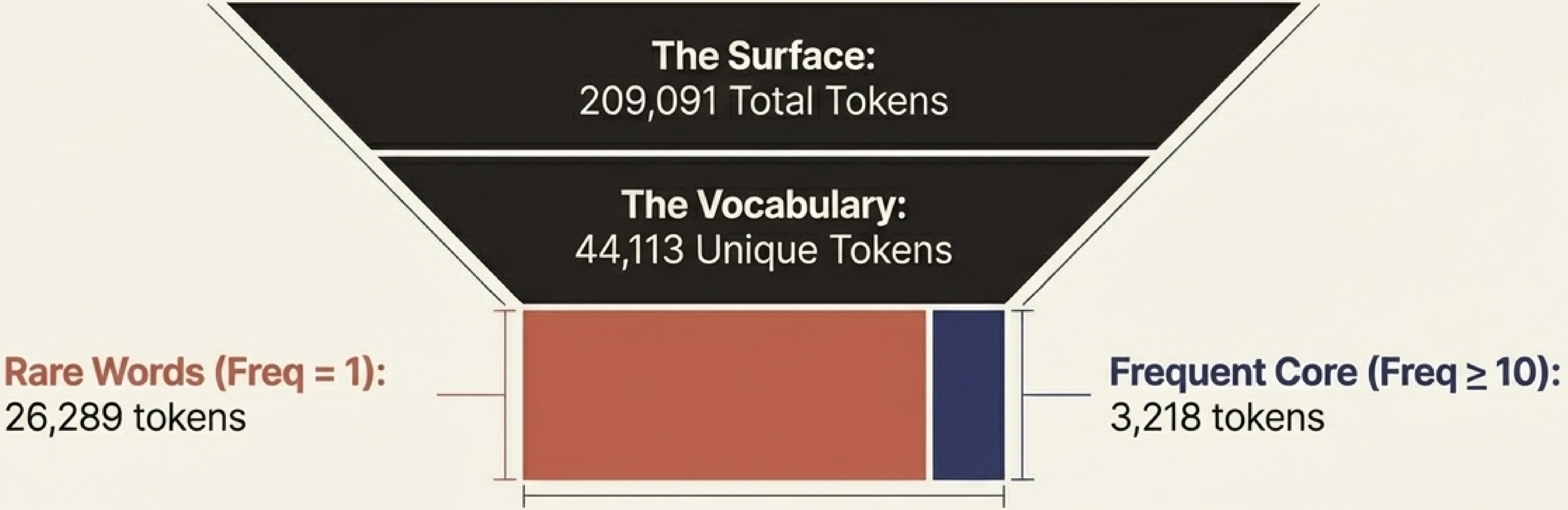


Total Documents:
1,058 cleaned poems (parquet format)

Authors:
10 distinct classes

The dataset was expanded from previous iterations by incorporating 211 new poems from the author Aşıq Ələsgər, introducing new stylistic variance into the multi-class sequence prediction task.

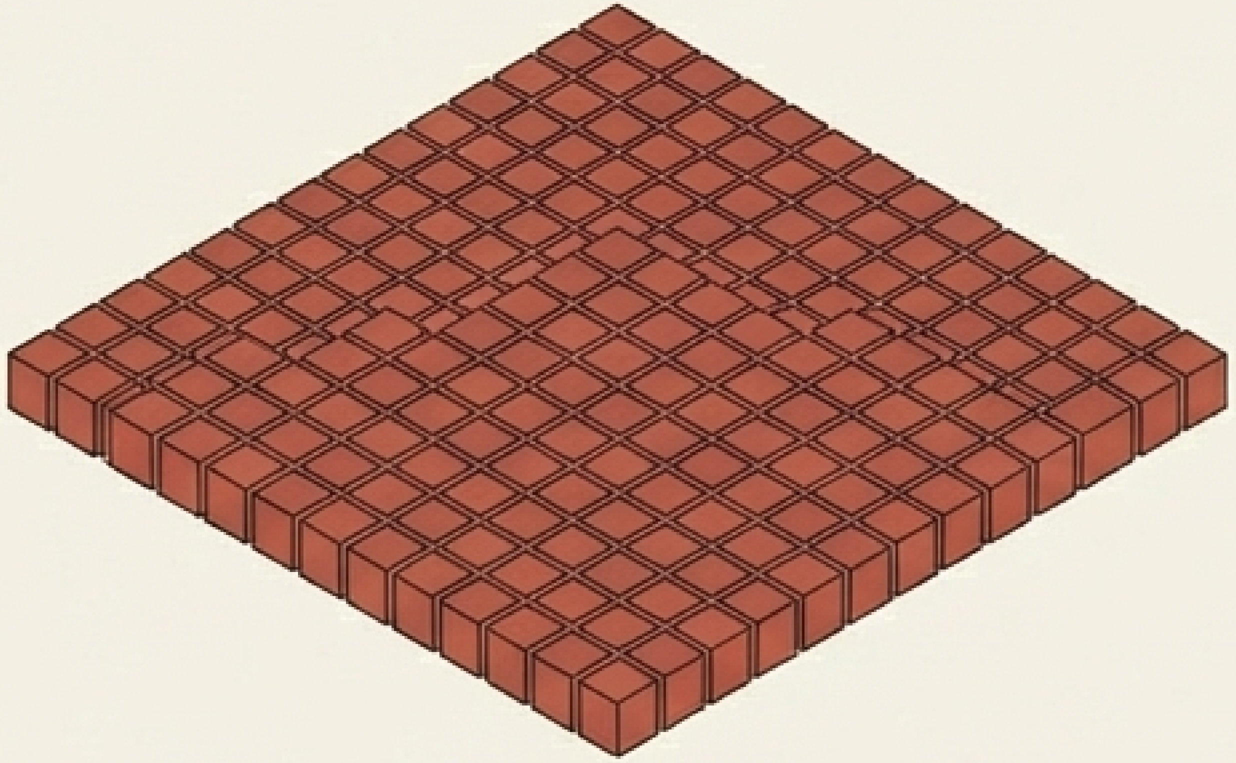
The Lexical Iceberg: Confronting Extreme Sparsity



Insight: The corpus exhibits a massive long-tail lexical pattern. With over 26,000 words appearing only once, frequency-thresholded representations are required to model stable co-occurrences.

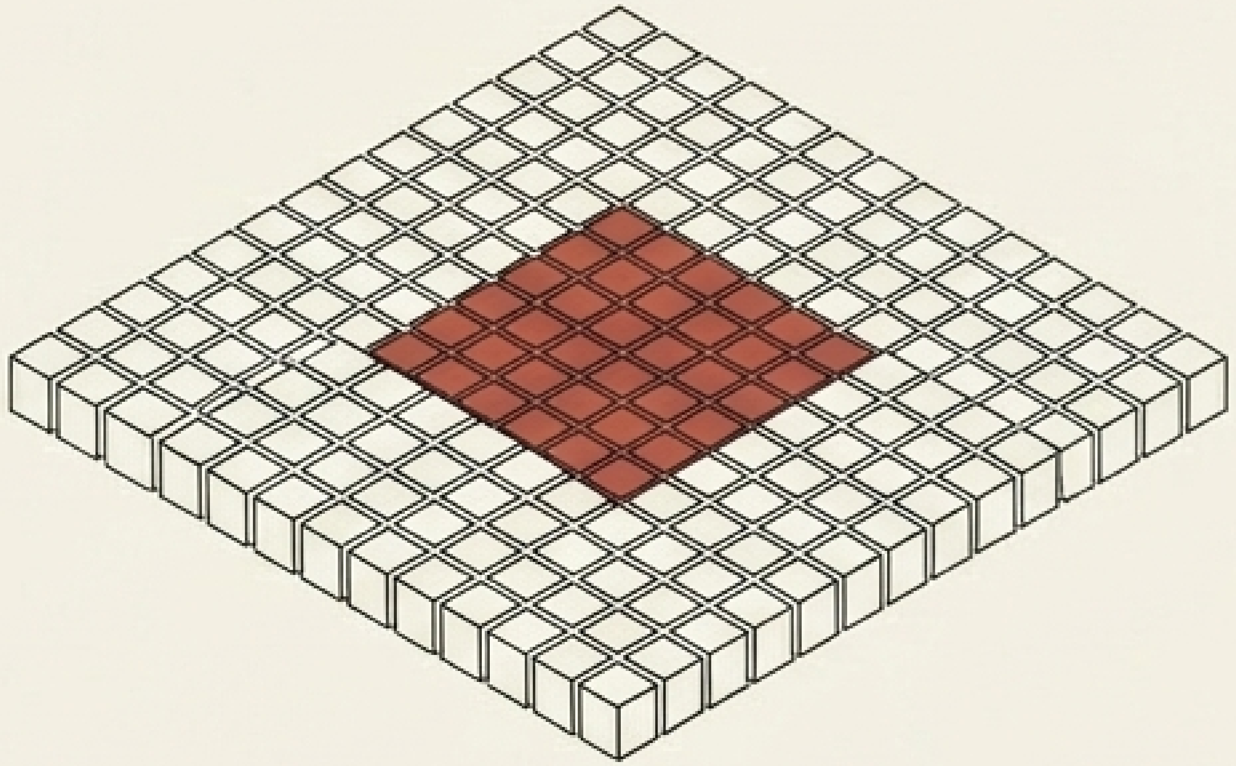
Structuring the Baseline Matrices

Matrix 1: Term-Document



Utilizes the full corpus vocabulary of 44,113 terms to capture global document occurrence.

Matrix 2: Word-Word



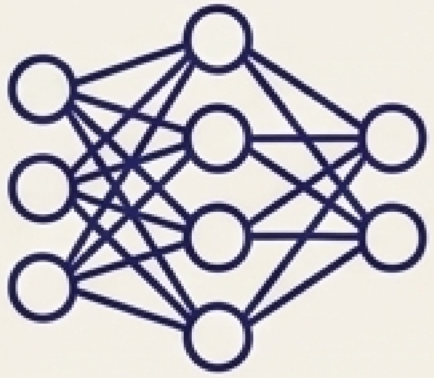
Restricted to the frequent core (tokens with freq ≥ 10) utilizing a tight context window of 2 to capture immediate local syntax and phrasing.

Artifacts Generated: Dense CSVs and Sparse NPZ matrices saved deterministically for pipeline ingestion.

The Dual Engines: Local Context vs. Global Co-occurrence

Shared Constraints | Embedding Dimension: 100 | Min Count: 5 | Window Size: 5 | Vocab Size: 6,575

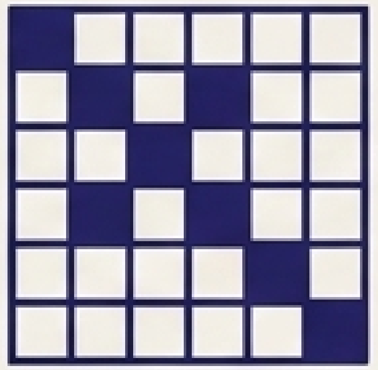
Word2Vec Engine



Architecture: **Skip-gram & CBOW**
Batch Size: 2048
Epochs: 8
Negative Samples: 5
Subsample: 0.0001

Final Loss: **CBOW** (1.32), **Skip-gram** (1.57)

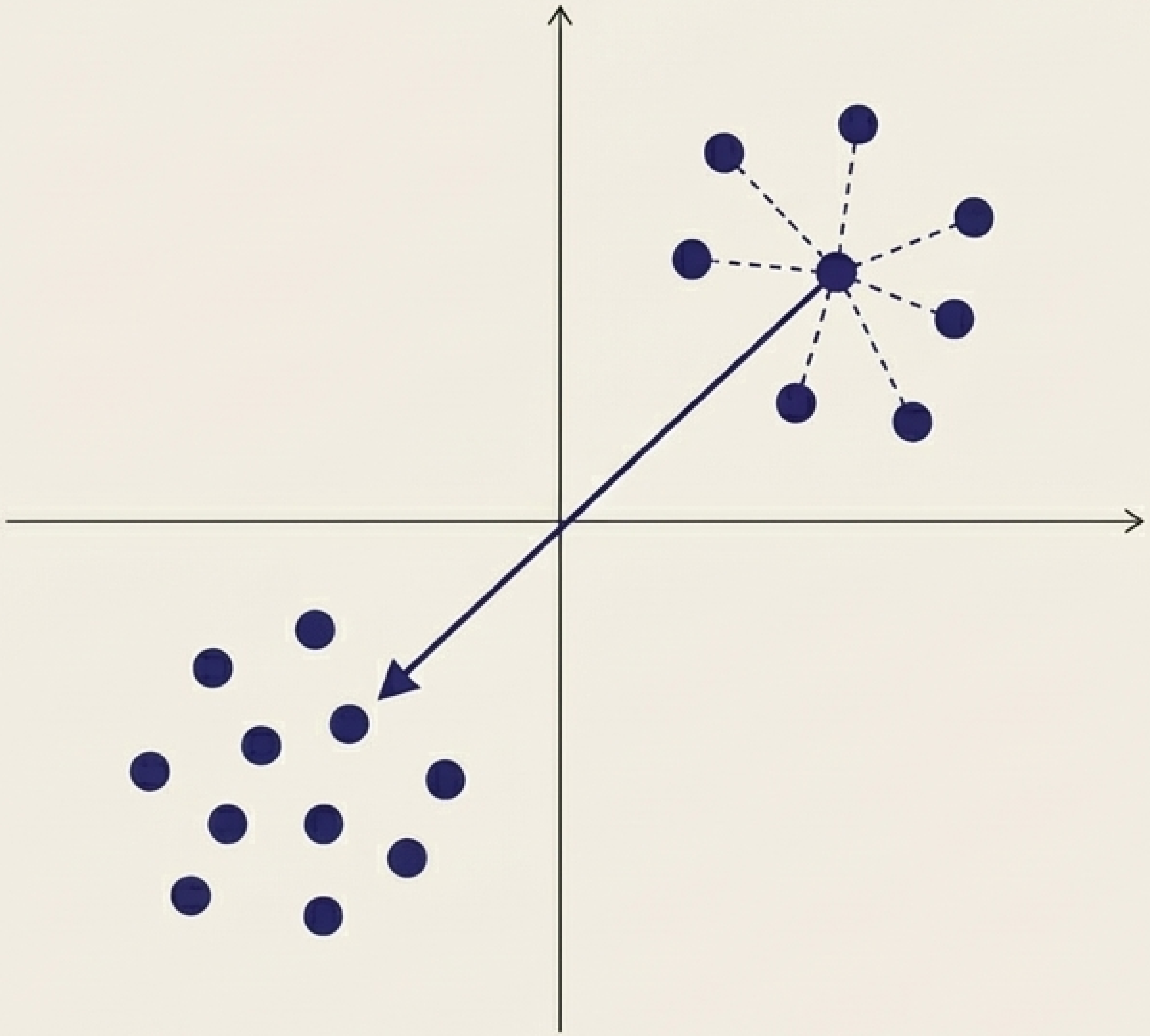
GloVe Engine



Architecture: **Global Log-Bilinear Regression**
Batch Size: 4096
Epochs: 25
x_max: 100.0
alpha: 0.75
Learning Rate: 0.05

Final Weighted Loss: 0.013

Probing the Semantic Spaces



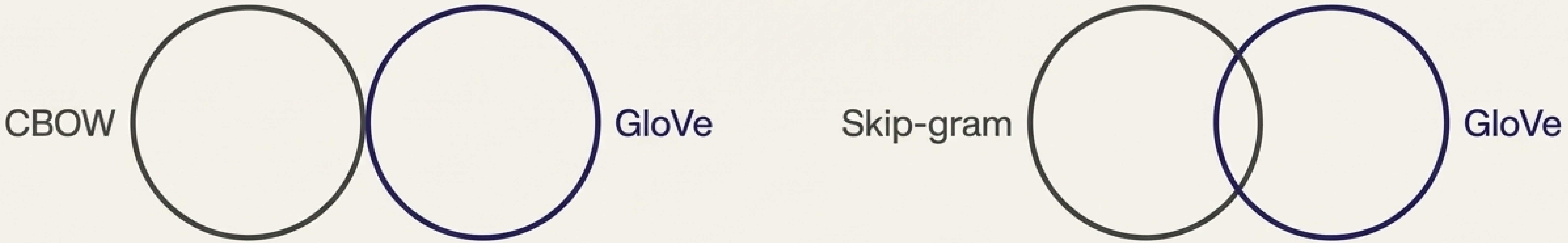
Insight 1: Nearest Neighbors

Both models successfully capture topical and stylistic cohesion unique to poetry. However, they map stylistic associations rather than strict, traditional synonymy.

Insight 2: Vector Arithmetic

Equation probes yield coherent but noisy analogical behavior. This noise is directly consistent with the highly specific and sparse contexts of poetic language.

Divergent Geometries: The Cross-Model Overlap

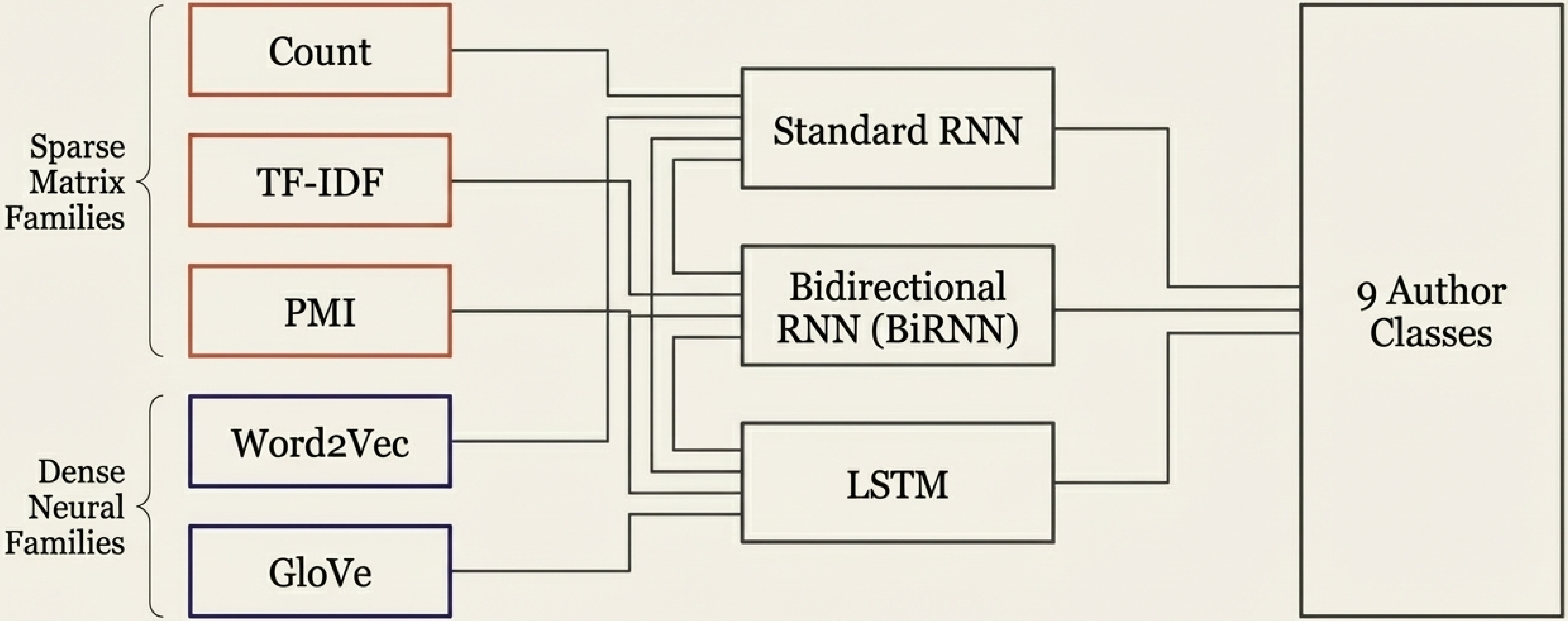


Word2Vec Model	Avg Jaccard Similarity	Avg Overlap Count	Equation Jaccard
CBOW	0.0269	0.50	0.0222
Skip-gram	0.0334	0.60	0.0000

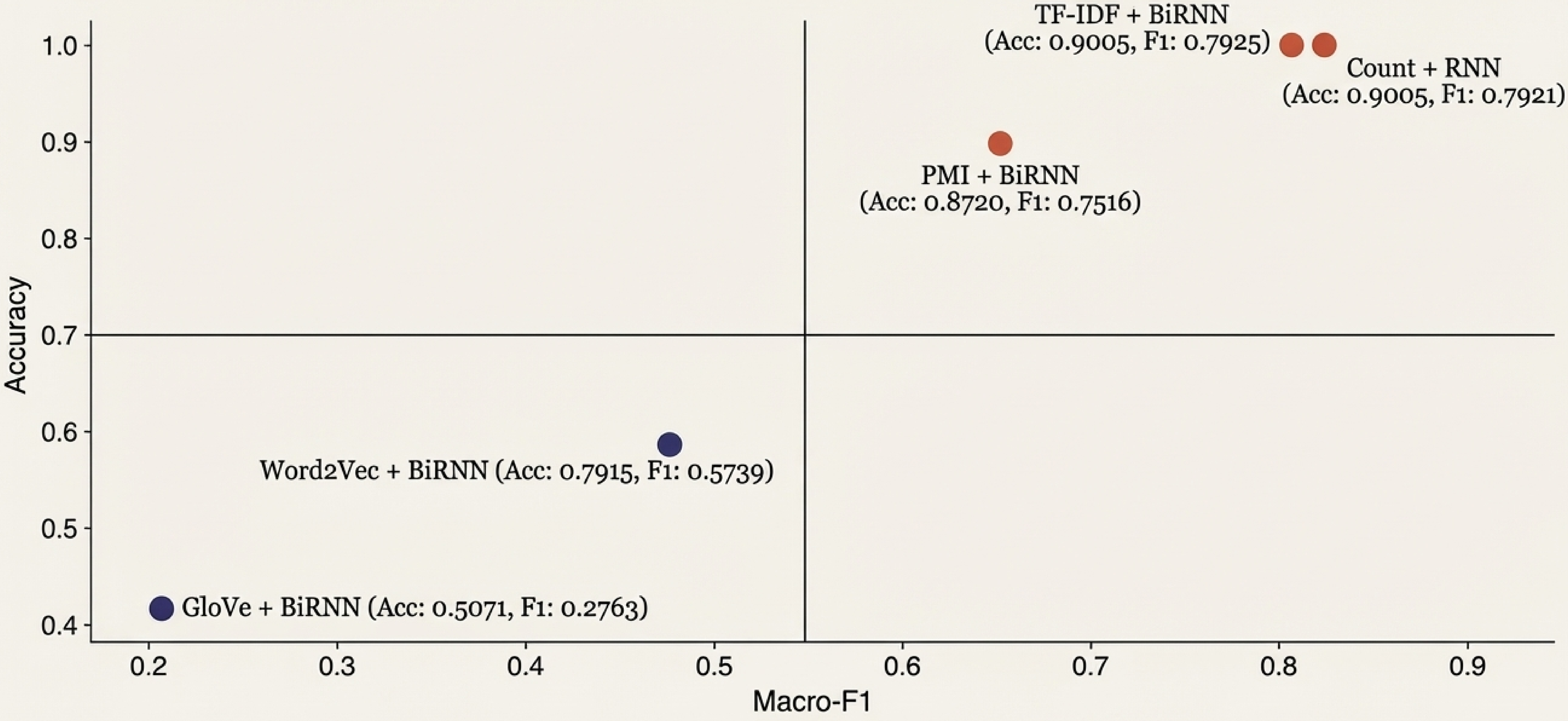
Insight: Cross-model agreement is staggeringly low. Skip-gram aligns slightly better than CBOW on neighbor overlap, but equation agreement is nearly zero. Word2Vec and GloVe encode completely different neighborhood geometries for the exact same poetic vocabulary.

The Classification Showdown Setup

Testing Parameters: 844 train documents, 211 test documents across 9 author classes.



The Performance Quadrant: Sparse Defeats Dense



Insight: The TF-IDF and Count features drastically outperform dense embedding-only features in macro-level predictive accuracy.

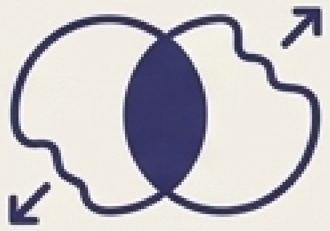
Synthesis: Why Did Count-Based Features Win?

The Foundation

Extreme Lexical Sparsity

The poetry dataset is fundamentally defined by over 26,000 rare words (Task 1).

The Distortion



Forced Dimensionality

When this sparse vocabulary is compressed into dense, low-dimensional spaces, it results in noisy, divergent neighborhood geometries (Task 4).

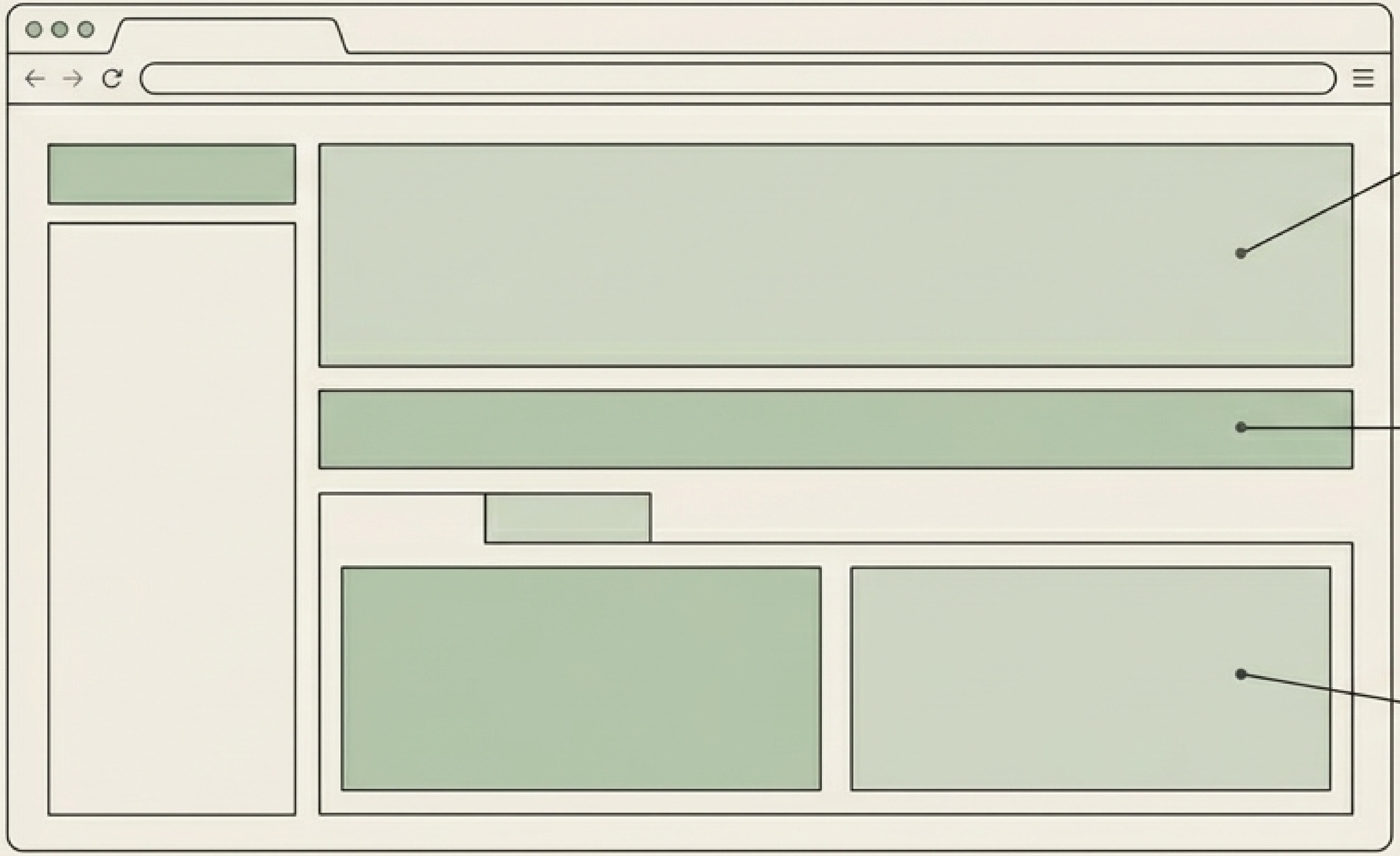
The Outcome

Critical Information Loss

Sequence models lack sufficient explicit stylistic supervision when utilizing these distorted embeddings.

The Conclusion: Simple, explicit TF-IDF features natively preserve the specific stylistic markers necessary for poetic author classification, effectively defeating current dense embedding pipelines.

The Integration Layer: FastAPI Command Center



Task 5 Bubble Chart: Interactive, sortable performance tables mapping the sparse vs dense showdown.

Animated Race View: Live visualization of average loss during training.

Task 4 Tabbed Comparison: User-input filtering for query words and model selection.

Insight: The Extra Task UI consolidates all pipeline outputs into a single, interpretable analysis layer, transforming static CSVs into an interactive, browser-based exploration tool.

Engineering Rigor & Reproducibility

```
$ python task5_rnn_classification.py
$ pytest
$ uvicorn app:app
```

Deterministic Artifacts

The entire blueprint is 100% reproducible. Core experiments operate on preserved deterministic settings (seed = 42 in training/config artifacts).

Verifiable Outputs

Every matrix, loss metric, and classification score can be regenerated directly from the project root.

Pipeline Review & Future Architectures

Summary

We successfully navigated from count-based corpus structures to neural embedding geometries and downstream sequence classification. We proved that for sparse poetic domains, explicit TF-IDF representations currently preserve stylistic data better than standard dense embeddings.

Future Vectors

Future pipeline iterations will focus on stronger contextual encoders (e.g., Transformers) and refined sequence input construction to better capture author-style discrimination directly within dense spaces.